

Research: Machine Learning Applications

Abstract

This study examines the effects of machine learning algorithms on business decision-making. We analyzed data from 200 companies over a 3-year period.

Introduction

The left column contains the introduction. Machine learning has transformed how businesses operate. Prior research has shown significant improvements in efficiency when ML systems are properly implemented. This paragraph continues in the left column so that the column has enough text to fill visibly.

Academic papers and newspapers often use two-column layouts. Text flows from the top of the left column to the bottom, then from the top of the right column to the bottom.

Methods and Results

In contrast, the right column contains different content. Multi-column PDFs cause PyPDF2 to read line-by-line across columns (wrong order): left line 1, right line 1, back to left line 2, and so on.

pdfplumber with layout detection reads the left column fully, then the right column (correct order). We conducted a comprehensive survey. Data was collected through interviews and questionnaires.

Conclusion

Results showed 35% improvement in decision-making speed when ML systems were used. Layout-aware extraction is essential for multi-column documents.